

1. Consider inserting the keys 14, 32, 35, 34, 15, 26, 18, 88, 51 into a hash table of length  $m = 11$  using open addressing with the auxiliary hash function  $h'(k) = k$ . Illustrate the result of inserting these keys using
- (a) linear probing,
  - (b) quadratic probing with  $c_1 = 1$  and  $c_2 = 3$ , and
  - (c) double hashing with  $h_1(k) = k$  and  $h_2(k) = 1 + (k \bmod (m - 1))$ .

**(a) Linear Probing**

|       |    |    |    |    |    |    |   |    |    |   |    |
|-------|----|----|----|----|----|----|---|----|----|---|----|
| Index | 0  | 1  | 2  | 3  | 4  | 5  | 6 | 7  | 8  | 9 | 10 |
| Value | 88 | 34 | 35 | 14 | 15 | 26 | - | 18 | 51 | - | 32 |

**(b) Quadratic Probing ( $c_1 = 1, c_2 = 3$ )**

|       |    |    |    |    |    |   |   |    |    |    |    |
|-------|----|----|----|----|----|---|---|----|----|----|----|
| Index | 0  | 1  | 2  | 3  | 4  | 5 | 6 | 7  | 8  | 9  | 10 |
| Value | 88 | 34 | 35 | 14 | 15 | - | - | 18 | 26 | 51 | 32 |

**(c) Double Hashing ( $h_1(k) = k, h_2(k) = 1 + (k \bmod 10)$ )**

|       |    |    |    |    |    |   |    |    |   |    |    |
|-------|----|----|----|----|----|---|----|----|---|----|----|
| Index | 0  | 1  | 2  | 3  | 4  | 5 | 6  | 7  | 8 | 9  | 10 |
| Value | 26 | 34 | 35 | 14 | 15 | - | 51 | 18 | - | 88 | 32 |

0 1 2 3 4 5 6 7 8

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- (a) linear probing,
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(a) Linear Probing  $\Rightarrow$  問隔壁 (idx  $\pm 1$ )

|     |    |    |    |    |            |            |   |            |            |   |    |
|-----|----|----|----|----|------------|------------|---|------------|------------|---|----|
| idx | 0  | 1  | 2  | 3  | 4          | 5          | 6 | 7          | 8          | 9 | 10 |
| val | 88 | 34 | 35 | 14 | 15         | 26         | / | 18         | 51         | / | 32 |
|     |    |    |    |    | $\uparrow$ | $\uparrow$ |   | $\uparrow$ | $\uparrow$ |   |    |
|     |    |    |    |    | 26         |            |   | 51         |            |   |    |
|     |    |    |    |    |            | $+1$       |   |            | $+1$       |   |    |

(b) Quadratic Probing  $\Rightarrow$  構造 2 次方程  $h(k, i) = (h'(k) + c_1 i + c_2 i^2)$   
 $i = -$  一直 try until 沒有碰撞

$$h(k, i) = (h'(k) + i + 3i^2) \bmod 11$$

|     |    |    |    |    |    |   |   |    |    |    |    |
|-----|----|----|----|----|----|---|---|----|----|----|----|
| idx | 0  | 1  | 2  | 3  | 4  | 5 | 6 | 7  | 8  | 9  | 10 |
| val | 88 | 34 | 35 | 14 | 15 | / | / | 18 | 26 | 51 | 32 |
|     |    |    |    |    |    |   |   |    |    |    |    |

$$i = 8: (51 + 8 + 192) \bmod 11 = 9$$

(c) Double hashing  $\Rightarrow$  用 2 次 hash function

$$h(k, i) = (h_1(k) + i h_2(k)) \bmod 11$$

$$h_1(k) = k \quad h_2(k) = 1 + (k \bmod 10)$$

|     |       |    |    |    |    |   |       |    |   |    |    |
|-----|-------|----|----|----|----|---|-------|----|---|----|----|
| idx | 0     | 1  | 2  | 3  | 4  | 5 | 6     | 7  | 8 | 9  | 10 |
| val | 26    | 34 | 35 | 14 | 15 | / | 51    | 18 | / | 88 | 32 |
|     | (i=1) |    |    |    |    |   | (i=6) |    |   |    |    |

$$h(14, i) = 14 + 5i \bmod 11$$

$$h(32, i) = 32 + 3i \bmod 11$$

$$h(35, i) = 35 + 6i \bmod 11$$

$$h(34, i) = 34 + 5i \bmod 11$$

$$h(15, i) = 15 + 6i \bmod 11$$

$$h(26, i) = 26 + 7i \bmod 11$$

$$h(18, i) = 18 + 9i \bmod 11$$

$$h(88, i) = 88 + 9i \bmod 11$$

$$h(51, i) = 51 + 2i \bmod 11$$